Introduction to Artificial Intelligence

Summer 2026

Project: Diamond price prediction

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This project is done in pairs.

1. Project details

This project aims to predict the price of diamonds. To ensure the quality of the model, it is important to compare the results with other solutions, conduct comprehensive methods research, and use suitable datasets for training and testing. The final output will be presented in paper form, including the methodology, datasets, and results of the developed model, and compared to existing solutions in the field.

There are many datasets for diamond price prediction, which can be utilized in this task. Proper documentation of dataset selection and usage should be included in the final report.

2. Useful references

1. PyTorch documentation: https://pytorch.org/docs/stable/index.html
2. Keras: https://www.tensorflow.org/guide/keras
3. Documentation and user guide from Scikit-Learn: https://scikit-learn.org/stable/user_guide.html
4. For getting started on the systematic process of Machine Learning projects, checkout Jason Brownlee's blog.

3. Datasets

1. https://www.kaggle.com/competitions/diamond-price-prediction
2. https://www.kaggle.com/search?q=diamond+in%3Adatasets

4. Technical details

1. The solution must be implemented in Python
2. Please ensure that your code adheres to basic standards of lean coding in accordance to PEP 8. Additionally, it should contain comments in the crucial parts to help with readability and understanding
3. The clear instruction how to run and test the code should be included
   The submission of the final report is mandatory, and the task will not be accepted without it

4. As part of the EARIN project requirements, a GitLab (our Faculty) repository should be created and maintained to store the Python code for the implemented solution. The repository will be considered an obligatory component of the project and will be used for version control, collaboration, and submission of the final code.

5. Handing-in guidelines

1. Project meetings will be conducted on MS Teams platform
2. The timeline for the project is as follows:
    i. Preliminary assumptions submission deadline is **May 15th**
    ii. The consultations for preliminary assumptions, which are scheduled for **May 18th**, will be arranged on an individual basis per pair
    iii. Midterm solution submission deadline is **May 29th.**
    iv. Midterm solution consultation will take place on **June 1st**, including the first project presentation.
    v. **The project submission deadline is June 12th**.
    vi. **Final solution presentation will take place on June 15th**.
3. For further information, please refer to the EARIN Project Instruction file.

6. Assessment Criteria

You can earn up to 25 points for your project, which will be evaluated based on the following criteria:

- 0-5 points: Preliminary documentation and consultations:
    - A short description of the algorithms that will be used, along with some examples.
    - Selection and description of the datasets.
    - Selection of the libraries
    - General plan of tests/experiments.
    - Methods of result visualization.
    - Definition of quality measures that will be used.
    - Submission of short presentation with all information above
    - Discussion during the consultation
- 0-5 points: Midterm solution documentation and consultations:
    - Submission of the current state of the Python code for the implemented solution in the GitHub repository.
    - Submission of the intermediate report to reflect the progress made so far, including:
        - problem description
        - proposed method description and introduction
        - intermediate results discussion and evaluation based on the proposed performance metrics
    - Discussion of intermediate results and analysis of the experiments conducted
    - Demonstration of the progress made during the project meetings with the instructor, including any intermediate results or findings.

    - Discussion of any challenges or issues encountered during the implementation and how they have been addressed.
    - Identification of any potential risks or obstacles that may affect the successful completion of the project

- 0-15 points: Final solution:
    - 0-5 points: Submission of the complete Python code for the implemented solution in the GitHub repository. Detailed instructions on how to run the code and reproduce the results.
        - Proper commenting and documentation within the code to ensure readability and understandability.
        - Adherence to good coding practices, including code organization, modularity, and efficiency.
    - 0-5 points: Detailed documentation in form of the chosen scientific paper template (IEEE, Springer, Nature, etc.)
        - Problem description
        - Dataset analysis
        - Related works research
        - Proposed method introduction and description
        - Final results discussion, assessment and comparison with other methods
        - Results visualizations
        - Conclusions
    - 0-5 points: Presentation of the final solution and results during the project presentation session.
        - Short presentation (it can have the structure from the final report). No more than 10 minutes.
        - Code demonstration:
            - Working training code (run for few epochs if it takes too long)
            - Inference of the pretrained and saved model checkpoint